

INS-02: Carrington Storm Motors — Actuarial Validation Framework

Document ID: CSM/INS/02 **Version:** 1.0 — Investor-Grade **Classification:** Confidential — For Qualified Investors and Insurance Partner Actuaries **Heuristic Foundation:** Accountant (mathematical rigor), Chester (probability economics), Williams (transparent assumptions) **Agents Consulted:** Insurance domain agent, Accountant, Chester, Zirconia (failure mode data) **Date:** July 2026

1.0 EXECUTIVE SUMMARY

The actuarial validation of Carrington Storm Motors’ loss reduction claims is the single most important gating item for the insurance integration business model. Before any insurer will reduce premiums for CSM-hardened assets, the insurer’s actuary must be convinced that the expected loss reduction is real, quantifiable, and durable.

This document presents CSM’s actuarial framework: the GIC-induced loss probability curves, transformer failure rate models with and without hardening, geographic correlation modeling, tail risk analysis, and capital relief calculations. It also outlines the 20 CSMinsurance dossiers (the subset of CSM’s 6,302 documents focused on insurance-specific actuarial analysis) and the external actuary engagement plan.

As the Accountant Heuristic demands: “The actuarial model is the bridge between materials science and insurance pricing. If the bridge collapses, the insurance model collapses with it. Build it strong.”

2.0 GIC-INDUCED LOSS PROBABILITY CURVES

2.1 Baseline GMD Frequency

Event Magnitude	Return Period	Probability per Year	Probability per Decade
G5 (Extreme)	~40-60 years	1.7-2.5%	17-25%
Carrington-class	~100-150 years	0.7-1.0%	7-10%
Carrington+ (“Black Swan”)	~500 years	0.2%	2%
Any G5 or greater	~40-60 years	~2%	~18-20%

Source: NOAA SWPC, National Academy of Sciences (2008), Lloyd’s of London (2013), JASON defense advisory group.

2.2 Transformer Failure Probability Curves (Without Hardening)

GMD Intensity (nT/min — dB/dt)	Transformer Failure Probability	Notes
<100 nT/min	<0.1%	Normal geomagnetic activity
100-300	0.1-1%	Minor storm — aging transformers at risk

GMD Intensity (nT/min — dB/dt)	Transformer Failure Probability	Notes
300-1,000	1-5%	Moderate storm — multiple trans-former heating events
1,000-3,000	5-20%	Major storm — significant GIC heating
3,000-5,000	20-50%	Severe storm — widespread trans-former damage
>5,000 (Carrington-level)	30-80%	Extreme — cascading trans-former failure likely

2.3 Transformer Failure Probability Curves (With CSM Charlemagne Hardening)

GMD Intensity (dB/dt)	Failure Probability (Hardened)	GIC Reduction	Failure Reduction
<100	<0.001%	N/A (no significant GIC)	N/A
100-300	<0.001%	99.97%	>99% reduction
300-1,000	<0.005%	99.97%	>99% reduction
1,000-3,000	<0.01%	99.97%	>99.5% reduction
3,000-5,000	<0.05%	99.97%	>99% reduction
>5,000	<0.1%	99.97% (beginning to saturate at extreme levels)	>98% reduction

Key Assumption: The 99.97% GIC reduction figure is based on TEM cell measurements with simulated GIC waveforms (see DD-01). At extreme GMD intensities (>5,000 nT/min), the shielding effectiveness may begin to compress (saturation effects in MXene). CSM's enhanced Charlemagne configuration (99.998% reduction) extends the effective range by approximately 20 dB.

2.4 Correlation Modeling Across Geographic Regions

GMD events are spatially correlated — a Carrington Event affects the entire planet simultaneously. However, GIC intensity varies by:

Factor	Correlation	Impact on Loss Modeling
Magnetic latitude	HIGH — GIC intensity increases with latitude	Northern US, Canada, Nordics, UK highest risk; equatorial regions lower risk
Ground conductivity	MODERATE — igneous rock regions (shield areas) experience higher GICs	Quebec, Scandinavia, Western Australia at elevated risk
Grid topology	MODERATE — long transmission lines in east-west orientation collect more GIC	US East-West interconnects; Canadian shield lines
Transformer design	LOW-MODERATE — older, single-phase, shell-form transformers more vulnerable	Asset-level risk varies

Implication for Insurance: GMD loss is highly correlated across large geographic regions. This is the opposite of what insurers prefer (uncorrelated risks). A single GMD event triggers claims across entire continents simultaneously — the classic “uninsurable” risk profile. CSM’s hardening decouples this correlation: hardened assets survive regardless of geography.

3.0 TAIL RISK ANALYSIS

3.1 The Insurance “Tail” Problem

The insurance industry models risk using probability distributions. “Tail risk” refers to the extreme end of the distribution — low-probability, high-consequence events. Carrington Events are the extreme tail: 1% annual probability, \$2-6 trillion consequence.

Traditional insurance pricing fails for extreme tail risks because: 1. Historical data is inadequate (one Carrington-level event in recorded history — 1859, pre-electric grid) 2. Premiums required to cover the tail would be unaffordable (\$20-60B annual expected loss → \$2,000-6,000/year per US household to fully price the risk) 3. Correlation destroys diversification benefits (every policyholder claims simultaneously)

3.2 How CSM Reshapes the Tail

Metric	Without CSM	With CSM (Hardened Grid)
Expected annual loss (US)	\$2-6B	\$0.02-0.6B (99% reduction)
99.5th percentile (1-in-200 year)	~\$2 trillion	~\$50-200B (90-97.5% reduction)
Tail fatness (kurtosis)	Very high (fat tail)	Significantly reduced (thinner tail)
Insurability	Borderline uninsurable (private market)	Insurable with CSM hardening requirements

CSM transforms GMD risk from an uninsurable tail risk into an insurable, manageable risk. This is the fundamental actuarial value proposition.

4.0 CAPITAL RELIEF CALCULATION

4.1 Solvency II — Standard Formula Impact

Under Solvency II (EU insurance regulation), insurers must hold capital against catastrophe risk. The Natural Catastrophe risk sub-module includes “other” perils, under which GMD would fall.

Scenario	Capital Requirement (per €1B of insured GMD exposure)
Without CSM hardening	€100-250M (10-25% capital charge — conservative, given fat tail)
With CSM hardening (loss reduction verified)	€10-50M (1-5% capital charge)
Capital relief	€90-200M per €1B exposure

4.2 NAIC RBC (US) — Risk-Based Capital Impact

Under the NAIC Risk-Based Capital framework for US insurers:

Scenario	R4 (Asset/Underwriting Risk) Charge
Without CSM hardening	Higher risk charge for concentrated GMD exposure
With CSM hardening	Reduced risk charge; potential for separate risk category (“hardened electrical infrastructure”)

Capital relief creates a direct financial incentive for insurers to require CSM hardening. The saved capital can be deployed to underwrite additional business, generating returns that far exceed the cost of the premium reduction offered to CSM-hardened policyholders.

5.0 THE 20 CSMINSURANCE DOSSIERS — SUMMARY

5.1 Dossier Framework

Each CSMinsurance dossier covers a specific insurance line or exposure:

Dossier #	Topic	Key Analysis
CSMINS-01	Homeowners Insurance — Aegis Home integration	Residential GMD loss model; premium reduction calculation
CSMINS-02	Commercial Property — Aegis Business integration	Commercial GMD loss model; business interruption
CSMINS-03	Equipment Breakdown — Charlemagne integration	Transformer failure model; replacement cost analysis
CSMINS-04	Business Interruption — time-element coverage	Grid-down business interruption duration model
CSMINS-05	Parametric GMD — trigger design	NOAA G5 alert trigger; payout structure; basis risk
CSMINS-06	Marine Insurance — CSM fleet integration	Vessel GMD navigation/communication loss model

Dossier #	Topic	Key Analysis
CSMINS-07	Aviation Insurance — ground systems hardening	Airport equipment GMD vulnerability model
CSMINS-08	Energy Insurance — utility property	Substation and transmission line GMD loss model
CSMINS-09	Reinsurance — GMD catastrophe treaty design	Aggregate GMD loss model; reinsurance layer attachment points
CSMINS-10	Reinsurance — capital relief analysis	Solvency II/NAIC RBC capital reduction
CSMINS-11	ILS / Cat Bond — CSM-triggered bonds	Insurance-linked security design; parametric trigger
CSMINS-12	Life Insurance — mortality impact	Excess mortality from prolonged grid failure (see ESG-02)
CSMINS-13	Health Insurance — medical system impact	Healthcare collapse impact on health claims
CSMINS-14	Workers' Compensation — workforce impact	Occupational injury during grid recovery
CSMINS-15	Auto Insurance — vehicle electronics	Auto electronics damage from EMP; claims modeling
CSMINS-16	Agriculture Insurance — food system impact	Crop/livestock loss from grid-dependent agriculture
CSMINS-17	Municipal Insurance — local government	Municipal service disruption claims
CSMINS-18	Cyber Insurance — electronic damage	Distinction between cyber attack and EMP damage
CSMINS-19	Supply Chain Insurance — contingent BI	Contingent business interruption from supplier grid failure
CSMINS-20	Political Risk Insurance — sovereign grid failure	Sovereign grid failure as political risk event

5.2 Dossier Maturity Levels

Maturity	Dossiers	Status
Level 1 — Framework Complete	CSMINS-01 through 05	Internal model; ready for actuary review
Level 2 — Data Populated	CSMINS-01, 02, 03, 05, 08, 09	Preliminary data from TEM cell testing
Level 3 — Pilot-Validated	None yet	Will require pilot program data (Year 2-3)
Level 4 — Industry-Standard	Target: CSMINS-01 through 05, 08, 09	Post-independent actuarial validation

6.0 EXTERNAL ACTUARY ENGAGEMENT PLAN

6.1 Why External Actuary?

Insurers will not accept CSM's actuarial models at face value. Independent actuarial validation is required from a firm with credibility in the insurance industry.

6.2 Target Actuarial Firms

Firm	Credibility	GMD/Climate Expertise
Milliman	HIGH — one of the largest actuarial consulting firms globally	Natural catastrophe modeling practice
Oliver Wyman (Actuarial)	HIGH	Catastrophe risk modeling; climate resilience
Willis Towers Watson (WTW)	HIGH	Catastrophe modeling; reinsurance broking; climate
Aon (Actuarial / Reinsurance Solutions)	HIGH	Cat modeling; impact forecasting
Guy Carpenter (Marsh McLennan)	HIGH	Reinsurance broking; cat modeling
RMS (Risk Management Solutions)	HIGH — cat modeling specialist	Physical risk modeling; acquired by Moody's
AIR Worldwide (Verisk)	HIGH — cat modeling specialist	Natural catastrophe models

6.3 Engagement Plan

Phase	Timing	Activity	Cost Estimate
Phase 1	Year 1	RFP to 3-5 actuarial firms; proposal review	\$50K-100K
Phase 2	Year 1-2	Model review engagement: actuarial firm reviews CSM's loss models, assumptions, and data; issues opinion letter	\$250K-500K
Phase 3	Year 2-4	Pilot validation: actuarial firm validates models against pilot program data	\$200K-500K
Phase 4	Year 4+	Ongoing model maintenance and recalibration	\$100K-250K/year

6.4 Deliverables

Deliverable	Description	Use
Actuarial Opinion Letter	Formal opinion on reasonableness of CSM's loss reduction estimates	Regulatory filings with state insurance departments
Model Validation Report	Detailed review of model assumptions, data quality, methodology	Insurer partner due diligence
Pilot Validation Report	Comparison of model predictions to pilot program actual data	Model refinement; regulatory confidence

Deliverable	Description	Use
Rate Filing Support	Actuarial justification for premium reduction in rate filings	State insurance department engagement

7.0 AGENT VOICES

Insurance Domain Agent: “The insurance industry’s relationship with catastrophe models is complicated. They trust models that have been validated by actual events (hurricane models after Katrina, earthquake models after Northridge). GMD models have never been validated — because the validation event hasn’t happened yet. CSM’s challenge is to build actuarial credibility without an actual Carrington Event. The path: transparent assumptions, conservative estimates, independent validation, and — critically — continuous monitoring data from hardened assets that shows GIC levels during minor storms are consistent with the model’s predictions.”

The Accountant (Precision): “The actuarial model is a financial model with lives on the line. An error in the model doesn’t just misprice insurance — it could leave transformers unprotected because the model underestimated the risk, or make protection unaffordable because the model overestimated the risk. The model must be conservative in its protection estimates (under-promise) but aggressive in its deployment (over-deliver). That’s the Accountant Heuristic applied to actuarial science.”

Mork (Probability Reality): “A 12% probability per decade is not a ‘low probability.’ It means that if you live to 80, there’s roughly a 65% chance you’ll experience a Carrington-level event in your lifetime. That’s not a black swan. That’s not a tail risk. That’s a reasonable expectation. The insurance industry should be pricing GMD risk like they price hurricane risk in Florida — as an inevitability, not a possibility. CSM’s actuarial models should force that conversation.”

Document prepared for qualified investors and insurance partner actuaries. All actuarial models are based on available space weather data, transformer failure studies, and CSM’s materials testing data. Models are subject to refinement as additional data becomes available. Independent actuarial validation is required before models can be used for insurance rate-making purposes.